

Introduction to FPGA Programming using Vivado

Hardware Lab

Installation

Version 2018.2 : <http://jatinga.iitg.ernet.in/download/>

XDC for Board: <https://github.com/Digilent/digilent-xdc/blob/master/Arty-A7-100-Master.xdc>

Part name: XC7A100TCSG324C-1 ([Nexys A7](#))

Issues on [Ubuntu](#):

- License:

```
$ export XILINXD_LICENSE_FILE=opt/licenses:2103@172.16.112.
```

- Drivers:

```
$ cd /etc/udev/rules.d
```

check for the following files, if not present, copy them and reboot:

- [52-xilinx-digilent-usb.rules](#)
- [52-xilinx-pcusb.rules](#)

Note: Some dependencies might need to be manually installed:

libncurses5-dev libncursesw5-dev libncurses5 libtinfo5 libtinfo-dev

Simple AND design

```
module and2 (a, b, c);  
    input a;  
    input b;  
    output c;  
    assign c = a & b;  
endmodule
```

Simple AND design testbench

```
`timescale 1ns / 1ps
module tb_and(); //no parameters in function in tb
    // Inputs are regs
    reg a;
    reg b;
    // outputs are wires
    wire c;
    // Instantiate the and2 module
    and2 uut (.a(a), .b(b), .c(c) );
    // Test procedure
    initial begin
        // Initialize inputs
        a = 0; b = 0; #10; // Test case 1: a = 0, b = 0
        $display("Test case 1: a = %b, b = %b, c = %b", a, b, c);

        a = 0; b = 1; #10; // Test case 2: a = 0, b = 1
        $display("Test case 2: a = %b, b = %b, c = %b", a, b, c);

        a = 1; b = 0; #10; // Test case 3: a = 1, b = 0
        $display("Test case 3: a = %b, b = %b, c = %b", a, b, c);

        a = 1; b = 1; #10; // Test case 4: a = 1, b = 1
        $display("Test case 4: a = %b, b = %b, c = %b", a, b, c);
        // End simulation
        $finish;
    end
endmodule
```

Resources

- YouTube Video: <https://www.youtube.com/watch?v=hKT1n0ixcJg>
- Manual for Board: <https://digilent.com/reference/programmable-logic/nexys-a7/start>